

**SECTION A**

**Multiple Choice Questions (1 Mark Each)**

**Q1.**

What is the mode of a data if median and mean are 9.6 and 10.5 respectively?

- (A) 7.8
- (B) 12.3
- (C) 8.4
- (D) 7

**Q2.**

Find the value of:

$$\left[ \tan A \operatorname{cosec} A - \sin A \sec A \right]^2$$

- (A) 0
- (B) 1
- (C) -1
- (D) 2

**Q3.**

A kite is flying at a height of 150 m from the ground. The string makes an angle of  $30^\circ$  with the horizontal. Find the length of the string.

- (A)  $100\sqrt{3}$  m
- (B) 300 m
- (C)  $150\sqrt{2}$  m
- (D)  $150\sqrt{3}$  m

**Q4.**

In triangles ABC and DEF,  $\angle B = \angle E$ ,  $\angle C = \angle F$  and  $AB = 3DE$ . Then the two triangles are:

- (A) Congruent but not similar
- (B) Congruent as well as similar
- (C) Neither congruent nor similar
- (D) Similar but not congruent

**Q5.**

If  $\theta$  is an acute angle and

$$\left[ \begin{array}{l} 7 + 4 \sin \theta = 9 \end{array} \right]$$

then  $\theta$  equals:

- (A)  $90^\circ$
- (B)  $30^\circ$
- (C)  $45^\circ$
- (D)  $60^\circ$

**Q6.**

From the given graph of two polynomials, find the number of distinct zeroes.

- (A) 3
- (B) 5
- (C) 2
- (D) 4

**Q7.**

In the given figure, PA is a tangent from external point P to a circle with centre O. If  $\angle POB = 115^\circ$ , find  $\angle APO$ .

- (A)  $25^\circ$
- (B)  $65^\circ$
- (C)  $90^\circ$
- (D)  $35^\circ$

**Q8.**

A wire 20 cm long is bent into an arc of a circle of radius  $\left(\frac{60}{\pi}\right)$  cm. Find the angle subtended at the centre.

- (A)  $30^\circ$
- (B)  $60^\circ$
- (C)  $90^\circ$
- (D)  $50^\circ$

**Q9.**

If  $\text{HCF}(98, 28) = m$  and  $\text{LCM}(98, 28) = n$ , find  $n - 7m$ .

- (A) 0
- (B) 28
- (C) 98
- (D) 198

**Q10.**

Which of the following is a rational number between  $(\sqrt{3})$  and  $(\sqrt{5})$ ?

**Q11.**

The sum of the zeroes of the polynomial

$$\begin{aligned} &[ \\ p(x) &= 5x - 7x^2 + 3 \\ &] \end{aligned}$$

is:

- (A)  $(-\frac{7}{5})$
- (B)  $(\frac{7}{5})$
- (C)  $(\frac{5}{7})$
- (D)  $(-\frac{5}{7})$

**Q12.**

If  $x = 1$  and  $y = 2$  satisfy

$$\begin{aligned} &[ \\ 2x - 3y + a &= 0 \\ &] \end{aligned}$$

and

$$\begin{aligned} &[ \\ 2x + 3y - b &= 0 \\ &] \end{aligned}$$

then:

- (A)  $a = 2b$
- (B)  $2a = b$
- (C)  $a + 2b = 0$
- (D)  $2a + b = 0$

**Q13.**

If the area of a sector is  $(40\pi)$  square units and the central angle is  $72^\circ$ , then the radius of the circle is:

- (A) 20 units
- (B) 10 units
- (C)  $(10\sqrt{2})$  units
- (D) 100 units

**Q14.**

The tangents drawn at the endpoints of a diameter are always:

- (A) Parallel
- (B) Perpendicular
- (C) Equal
- (D) Intersecting

**Q15.**

If

$$\begin{bmatrix} (-1)^n + (-1)^8 = 0 \end{bmatrix}$$

then  $n$  is:

- (A) Any positive integer
- (B) Any negative integer
- (C) Any odd integer
- (D) Any even integer

**Q16.**

The endpoints of a diameter are  $(2, 4)$  and  $(-3, -1)$ . Find the radius.

**Q17.**

The 11th and 13th terms of an AP are 39 and 45 respectively. Find the common difference.

- (A) 42
- (B) 21
- (C) 6
- (D) 3

**Q18.**

A card is drawn at random from a deck of 52 cards. Find the probability that the card is a spade or a king.

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### Assertion–Reason Questions

**Q19.**

**Assertion (A):** The probability of selecting a number at random from the numbers 1 to 20 is 1.

**Reason (R):** If  $P(E)=1$ , then E is called a sure event.

**Q20.**

**Assertion (A):** If two hemispheres of the same radius are joined along their bases, a sphere is obtained.

**Reason (R):** Total surface area of a sphere of radius r is  $(3\pi r^2)$ .

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### SECTION B

#### Very Short Answer Questions (2 Marks Each)

**Q21. (a)**

If

$$\left[ \begin{array}{l} \triangle ABC \sim \triangle PQR \end{array} \right]$$

where  $AB = 6$  cm,  $BC = 4$  cm,  $AC = 8$  cm and  $PR = 6$  cm, find  $PQ + QR$ .

**OR**

**Q21. (b)**

Given:

$$\left[ \frac{QR}{QS} = \frac{QT}{PR} \right]$$

and  $\angle 1 = \angle 2$ ,

show that:

$$\left[ \triangle PQS \sim \triangle TQR \right]$$

**Q22. (a)**

If

$$\left[ x \cos 60^\circ + y \cos 0^\circ + \sin 30^\circ - \cot 45^\circ = 5 \right]$$

find  $x + 2y$ .

**OR**

**Q22. (b)**

Evaluate:

$$\left[ \frac{\tan^2 60^\circ}{\sin^2 60^\circ + \cos^2 30^\circ} \right]$$

**Q23.**

A person stands at point P outside a circular ground at a distance of 26 m from its centre. PA and PB are tangents of length 10 m each. Find the radius of the circular ground.

**Q24.**

Find the zeroes of:

$$\left[ p(x) = x^2 + \frac{43}{4}x - \frac{43}{4} \right]$$

**Q25.**

In triangle ABC with

$A(9, -2)$ ,  $B(-3, 7)$ ,  $C(-1, 10)$ ,

find the length of the median drawn from vertex B.

**SECTION C****Short Answer Questions (3 Marks Each)****Q26.**

Prove that  $(\sqrt{5})$  is irrational.

**Q27.**

Two dice are rolled together. Find the probability of:

- (i) A multiple of 2 on one die and a multiple of 3 on the other die.
- (ii) Product of the numbers obtained being a perfect square.

**Q28. (a)**

Prove:

$$\left[ \frac{\tan \theta}{1 - \cot \theta} + \frac{\cot \theta}{1 - \tan \theta} \right] \\ 1 + \sec \theta \operatorname{cosec} \theta$$

**OR**

**Q28. (b)**

Prove:

$$\left[ \frac{\sin A + \cos A}{\sin A - \cos A} + \frac{\sin A - \cos A}{\sin A + \cos A} \right] \\ \frac{2}{2 \sin^2 A - 1}$$

**Q29.**

A room consists of a cylinder surmounted by a hemisphere. The radius of the hemisphere is half the height of the cylindrical portion. If the volume of air in the room is

$$\left[ \frac{1408}{21} \text{m}^3 \right]$$

find the height of the cylindrical portion.

**Q30. (a)**

In the given figure, O is the centre of the circle and BCD is a tangent at C.

Prove that:

$$[\angle BAC + \angle ACD = 90^\circ]$$

**OR**

**Q30. (b)**

Prove that opposite sides of a quadrilateral circumscribing a circle subtend supplementary angles at the centre.

**Q31.**

Find the ratio in which the y-axis divides the line segment joining:

$$[(5, -6) \text{ and } (-1, -4)]$$

Also find the point of intersection.

## SECTION D

### Long Answer Questions (5 Marks Each)

**Q32. (a)**

The perimeter of a right-angled triangle is 60 cm and its hypotenuse is 25 cm. Find the lengths of the other two sides.

**OR**

**Q32. (b)**

A train covers a distance of 480 km at a uniform speed. If its speed were 8 km/h less, it would take 3 hours more. Find the speed of the train.

**Q33.**

A bag contains red and blue balls.

10% of red balls plus 20% of blue balls equals 24.

Three times the number of red balls exceeds the number of blue balls by 20.

Find the number of red and blue balls.

**Q34.**

The lengths of 40 leaves were measured and grouped as follows:

**Length (mm) Frequency**

|         |    |
|---------|----|
| 118–126 | 3  |
| 127–135 | 5  |
| 136–144 | 9  |
| 145–153 | 12 |
| 154–162 | 5  |
| 163–171 | 4  |
| 172–180 | 2  |

Find the median length of the leaves.

**Q35. (a)**

The diagonal BD of parallelogram ABCD intersects AE at F where E lies on BC.

Prove that:

$$\begin{aligned} &[ \\ &DF \times EF = FB \times FA \\ &] \end{aligned}$$

**OR**

**Q35. (b)**

In  $\triangle ABC$ , if

$$\begin{aligned} &[ \\ &AD \perp BC \\ &] \end{aligned}$$

and

$$\begin{aligned} &[ \\ &AD^2 = BD \times DC \\ &] \end{aligned}$$

prove that:

$$\begin{aligned} &[ \\ &\angle BAC = 90^\circ \\ &] \end{aligned}$$

